

Fixed gauge feedback and the Goldstone Ward boundary

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The new result

The protected flavor action now has a fixed-coupling electroweak-vector calculation and a solved local branch of its explicitly calibrated hard one-loop potential. The calculation adds the W and Z determinants to all 49 neutral scalar modes and the declared quark blocks. It distinguishes a genuine fixed-gauge-coupling test from the preceding continuation that scaled every Yukawa and scalar interaction. Three executed weak points retain positive hard-potential curvature and nonzero canonical weak CP after a single Higgs mass input is calibrated. The gauge contribution does not introduce a direct relative-source-frame force.

This is a result about a specified truncated potential. Its radial gauge correction is much larger than the retained radial tree curvature. Consequently, numerical positivity cannot be promoted to a uniformly controlled physical vacuum, a Higgs pole mass, a prediction of the electroweak scale or a golden CKM prediction. The finite Higgs mass adjustment and the gauge couplings are explicit inputs. The calculation identifies the reorganization required for the next physically matched action rather than hiding that issue inside a neutral-only weak limit.

Gauge inputs and exact pullback

The only electroweak-charged scalar in the declared action is one canonical Higgs doublet, with $H_{\text{dagger}}H = h^2/2$. Every family source and finite mediator is an electroweak singlet. Take $g=0.65$ and $g_{\text{prime}}=0.36$ as declared numerical inputs at MS scale $\mu=1$ in the model's dimensionless units. These are benchmark inputs, not a scale-matched extraction from experimental W, Z or Higgs data. Lepton contributions, gauge anomaly completion, momentum-dependent observables and running are not inferred from these numbers.

For the physical Higgs radius x , the vector squared masses are $a_W x^2$ and $a_Z x^2$, where $a_W = g^2/4$ and $a_Z = (g^2 + g_{\text{prime}}^2)/4$. The Landau-gauge MS vector potential is the sum $n a^2 x^4 [\log(a x^2/\mu^2) - 5/6]/(64 \pi^2)$, with multiplicities $n_W=6$ and $n_Z=3$. The photon and gluons have zero background-dependent masses and supply no one-loop vacuum determinant here. The vector subtraction is $5/6$, not the scalar/fermion subtraction $3/2$. Its physical-radius force is the sum $n a^2 x^3 [\log(a x^2/\mu^2) - 1/3]/(16 \pi^2)$, and its curvature is the sum $n a^2 x^2 [3 \log(a x^2/\mu^2) + 1]/(16 \pi^2)$. Tests independently differentiate the potential and force.

In the preceding continuation $V_r(X) = V_1(\sqrt{r}X)/r$, write $Y = \sqrt{r}X$. Keeping the gauge couplings fixed gives the pulled vector potential $U_g(Y_h) = r V_g(Y_h/\sqrt{r})$. This is the sum $n a^2 Y_h^4 [\log(a Y_h^2/(r \mu^2)) - 5/6]/(64 \pi^2 r)$. Its force therefore grows generically as $\log(1/r)/r$ at fixed Y_h . It does not inherit the r suppression of the scalar-plus-quark force. Symbolic differentiation proves the pullback, the calibrated curvature and the scale derivative. The latter is $-\sum n a^2 Y_h^4/(32 \pi^2 r)$, a pure Higgs quartic contact.

At $r=0.000001$ the unperturbed physical Higgs radius is approximately 42.4264, with vector squared masses 190.125 and 248.445 in the declared units. The pulled Higgs force is about 0.302312 and its uncalibrated curvature is about 24.2057. The raw tree-Hessian linear response has relative full-background displacement about 297.859 and a Higgs displacement about -15225.6 times its reference radius. Its displaced neutral scalar Hessian is not positive. These are failed linear diagnostics, not an asserted new vacuum. They establish that the neutral-only continuation cannot be reused at fixed gauge couplings without a separate electroweak radial treatment.

A finite Higgs mass condition and its local branch

Choose a reference radius h_0 equal to the rescaled tree value and add the finite quadratic contact $-\delta_{\mu^2} h^2/2$, where $\delta_{\mu^2} = U_g'(h_0)/h_0$. This is a Higgs tadpole calibration. It does not fit a quark mass, source angle, quartet or radial curvature. The contact exactly cancels the direct vector tadpole at the reference point. At $r=0.000001$ its coefficient is about 7.12556627 in physical mass-squared units. The scalar and quark hard terms still determine the residual coupled displacement.

The remaining direct vector curvature at h_0 is the sum $n a^2 h_0^2 [\log(a h_0^2 / (r \mu^2)) + 2/3] / (8 \pi^2 r)$. It is about 17.0801 at the solved nearby branch and about 36493.7 times the reference neutral tree h - h entry. A mass calibration therefore does not restore the old tree spectrum: a radial radiative hierarchy remains. The implementation evaluates the calibrated force with log ratios and factored differences near h_0 , avoiding subtraction of nearly equal large tadpoles. The curvature has not been subtracted or independently tuned.

Solve $\text{grad}[V_{\text{tree}}(Y) + r V_{\text{CW},49\text{-neutral}+\text{quarks}}(Y) + U_g(Y_h) - \delta_\mu^2 Y_h^2 / 2] = 0$. The finite contact is outside the tree propagators in this declared one-loop truncation. Updates include the direct calibrated vector curvature, and a domain guard rejects steps leaving the positive neutral bare spectrum. At $r=0.000001$ the solver converges after two updates with maximum tadpole residual about 1.20 times ten to the minus fifteen. The smallest bare neutral eigenvalue is about 9.64 times ten to the minus six. The smallest curvature of the hard potential is about 1.12549 times ten to the minus five. The relative full-background displacement is about 5.90648 times ten to the minus six, and the relative Higgs-radius change is about -6.00 times ten to the minus nine.

Canonical quark matching at this background retains a nonzero quartet J approximately 0.0002620084803. These remain tree masses and currents evaluated at a loop-selected background, not fully matched loop observables. Two further runs at $r=0.0000003$ and $r=0.0000001$ reproduce positive bare and hard curvatures and nonzero CP. This is continuation evidence for the stated truncated action, not interval-certified existence or global selection. The dominant radial correction means a Coleman-Weinberg-style power counting, consistently dressed scalar propagators and matching must be established before presenting this family as uniformly perturbative.

What the gauge contribution can select

The direct vector potential is a function of the Higgs radius alone. Each of the eight quadratic source-Higgs portals depends only on the sector singlets and $\text{Tr}(A^2)$. Under independent source conjugations $A \rightarrow U A U^\dagger$ these portals are constant. Infinitesimally, the norm variation is proportional to $\text{Tr}(A[B,A])=0$ by cyclicity. Thus the direct vector force has exactly zero projection on all twelve isospectral source directions. Profiling the Higgs in the vector-plus-quadratic-portal subsystem cannot introduce relative-frame dependence on a chosen radial branch.

This excludes that subsystem as the missing golden-frame selector. It does not exclude indirect changes of the full finite-family vacuum: the existing scalar interactions can correlate radial and angular displacements through their Hessian. The raw response records that induced angular displacement separately from the exactly zero direct gauge torque. Mixed gauge-scalar diagrams at higher loop order also lie beyond this one-loop statement. The required frame correlation must still come from justified finite-family source contractions and their quark couplings; adding the vector determinant alone does not supply it.

Goldstone organization and the Ward diagnostic

Ordinary Landau-gauge Goldstone logarithms produce infrared problems when evaluated near a symmetry-breaking minimum. We use the one-loop zero rule of the bare anti-resummation organization described by Stephen Martin in August 2026: Goldstone propagators are massless, squared masses are loop-order two-point insertions, and pure one-loop Goldstone vacuum integrals are scaleless in dimensional regularization. At this order they contribute zero. This implements a specific one-loop organization, not the higher-loop insertion rules or a computed momentum-dependent Goldstone self-energy.

For an $O(4)$ -invariant single-doublet radial potential, the angular Ward curvature is the radial tadpole divided by h . The implementation records the tree-plus-finite-contact contribution $G_{\text{tree}} = V_{\text{tree},h} / h - \delta_\mu^2$ and the declared hard correction $G_{\text{hard}} = r V_{\text{CW},h} / h + U_g, h / h$. Their sum equals the full hard tadpole divided by h . At the first solved branch they are approximately -7.12556627191 and +7.12556627191, cancelling at numerical precision. The cancellation is independently checked through the recomputed tadpoles. It is an algebraic Ward test of this radial potential, not independent proof of a Goldstone pole mass or of gauge-independent observables.

This cancellation also makes the power-counting limitation concrete. The mass adjustment is large relative to the original soft radial tree sector. Inserting it into the scalar propagators without reorganizing the approximation can invalidate the positive bare-spectrum calculation used by the hard loop determinant. Treating it as a finite one-loop contact defines the truncated calculation unambiguously, but does not remove that consistency obligation. A consistent electroweak-matched model needs a radial power counting and mass/current matching that carry these contacts and the Goldstone organization together.

Reproduction and references

Run `PYTHONPATH=python OPENBLAS_NUM_THREADS=1 python python/develop_flavor_gauge_feedback.py`. The scientific receipt contains four raw fixed-gauge continuation diagnostics, three calibrated local branches, exact identities, all input values, canonical quark spectra and Ward residuals. The focused tests check vector multiplicities, the 5/6 subtraction, physical-coordinate pullback, independent derivatives, calibration, stable near-reference evaluation, all quadratic portal tangents, independently recomputed branch tadpoles and a finite-gradient Higgs curvature column.

Stephen P. Martin, Two-loop effective potential for a general renormalizable theory and softly broken supersymmetry, Phys. Rev. D 65, 116003 (2002), <https://arxiv.org/abs/hep-ph/0111209>, provides the Landau-MS one-loop vector convention, especially equation 3.4. Stephen P. Martin, Taming the Goldstone contributions to the effective potential, Phys. Rev. D 90, 016013 (2014), <https://arxiv.org/abs/1406.2355>, describes the infrared and resummation problem. Stephen P. Martin, Bare effective potential and Goldstone boson anti-resummation, arXiv:2608.04182 (2026), <https://arxiv.org/abs/2608.04182>, gives the massless-propagator organization and the one-loop zero rule in section III. Our implementation uses those one-loop conventions; it does not implement the papers' multi-loop Standard Model results.